RESUME

Youngdu Min · Unity Game Client Programmer

dudenx02@gmail.com · Incheon, Korea · GitHub · Portfolio · LinkedIn

Unity client programmer with 3 years of live-service experience shipping "ToctocMath," a KERIS digital-textbook-linked math education app, to iOS, Android and WebGL, taking the lead on the team's game features such as mini-games. Contributed to 167% YoY download growth and the 1st-place 2025 Korea Quality Satisfaction Award by improving the service around teachers' needs and optimizing the app, including a ~70% build-size reduction. Shipped 5+ games as side projects, including a Steam title built as lead programmer on a team of 7. Since leaving, I have been strengthening CS fundamentals through a Computer Science degree and earned the Engineer Information Processing certification.

Core Skills

- **Unity**: Unity 6 / URP, UGUI development & optimization, ScriptableObject architecture, Addressables memory management, NavMesh, Timeline, DOTween, UniTask
- **C# & Architecture**: OOP / SOLID; Observer, Strategy, Factory and FSM patterns; legacy code refactoring and library development
- **Gameplay**: Corgi Engine AI extension (9 enemy types, 5 bosses) and FSM-based AI; weapon, skill, dialogue, quest and interaction systems; object pooling; localization (Korean / English)
- **Procedural Generation**: MST, A* and Union-Find map generation pipeline; seed-based deterministic generation
- **Optimization & Multi-platform**: Unity Profiler, build size cut from 1 GB+ to under 350 MB (~70%), WebGL memory-leak fixes, live operation on iOS / Android / WebGL
- **Networking**: FishNet server-authoritative design (SyncVar / RPC, late-join sync), seed sync with no map-data transfer, JSON / REST API, WebGL ↔ JavaScript interop
- **Release**: Steam, App Store and Google Play build, review and release; WebGL verified on a test server before CDN deployment
- **Tools & Collaboration**: Custom editor tools (CSV export for reports, prefab finder by root/component, batch texture resizing, prefab wrapper generation); led Git workflow; dev manuals; Jira, Trello, Notion, Obsidian, Figma, Adobe XD, Slack
- **AI-assisted Development**: Claude Code and other AI coding tools for design, implementation and debugging, with manual verification

Experience

Cemware · Unity Developer · Aug 2022 – Jul 2025 (3 years, incl. alternative military service)

ToctocMath · KERIS digital-textbook-linked math education app (incl. mini-games), live service · 8 developers (up to 12 with interns), lead programmer · iOS / Android / WebGL

- Took the lead on the team's game features: built 8+ casual educational mini-games and reworked most existing ones for curriculum changes and live issues, on a shared architecture of abstract classes and interface-based composition with data-driven settings
- Contributed to 167% YoY download growth and the 1st-place 2025 Korea Quality Satisfaction Award (Elementary Math) by improving core systems around teachers' needs and optimizing app size and performance
- Cut build size from over 1 GB to under 350 MB (~70%) across all platforms by replacing assets with reusable sliced UI sprites, removing unused assets, and building an editor tool for batch texture compression; fixed WebGL memory leaks and WebGL loading failures on Chromebooks and Whalebooks used in schools
- Integrated LMS embedding and AI backend communication (learning analytics, problem recommendation)
- Refactored legacy code into modular components; led Git workflow and releases (store review, WebGL CDN deployment); wrote and shared dev manuals

Shipped Games

Hollow: Cook Off · Side project · 2D action side-scroller (Corgi Engine; 12 stages + 5 boss stages) · Lead programmer, team of 7 · 2024 · Steam · Trailer · Feature demo · Code

- Extended Corgi Engine's AI Brain with boss-phase management and State/Decision tracking & recovery
- Derived many custom AI Actions and Decisions to build AI for 9 enemy types and 5 multi-phase bosses (randomized attack-pattern groups per phase, a healer AI that repairs allies' armor, etc.); built an event system
- Implemented weapons and abilities the engine did not support: bullet parry (reflect enemy bullets with slow motion), grappling gun, boomerang
- Built a JSON-based dialogue system, Korean/English localization, a save system and Timeline cutscenes
- Owned the full Steam release pipeline: build, review and release

Karma Defence · Defense · Team of 5, sole programmer · 2020 – 2021 · Google Play (service ended) · Code

- Built all programming, including multi-type unit object pooling (pool doubles when exhausted), stage quests with star rewards, upgrades, and a conditional inspector tool for designers

Others

- **Keyboard Invader** (team of 3, lead programmer, 2021): 2D shooter where each key fires a unique bullet; built a 2-phase boss AI and the shop; Excellence Award (2nd place), SBCF 2021 Game Category · Video · Play
- **Gun Slash** (solo, itch.io 2021, Corgi Engine): Local multiplayer shooter for up to 4 players with a lobby and custom input system supporting 2 keyboards + 4 gamepads · Video · Play
- **Gravity Miner** (university team of 7, main programmer, 2019 – 2020): Built core gameplay and physics for an arcade platformer with speed-based stomp combat and unique block gimmicks · Video · Play
- Game jams (3 submissions): Smilegate Beaver Jam 2023 Option Adventure (lead programmer), GMTK Game Jam 2022 Dice Knight and 2020 Dungeon Keyboarder

Technical Projects

Asymmetric Multiplayer Horror (Backrooms concept) · Unity 6 / URP / FishNet · Side project, team of 4 · Mar 2026 – Oct 2026 (discontinued after prototyping) · Built with AI coding tools (Claude Code) · Map generation video

- Owned procedural map generation: designed a 6-stage pipeline (room placement → Prim MST → cost-weighted A* corridors → loop / dead-end generation and cleanup → start/goal placement → validation); the server syncs only the seed and each client generates the same map locally
- 3D modular placement: editor tool that parses kit prefab names to auto-generate 73 metadata wrappers; placement hashes verify host/client consistency
- Server-authoritative hold interactions (occupancy lock, item validation/consumption, progress extrapolation), escape loop, and a player-controlled entity's dash-attack FSM and tracking system

Unit System for Unity · Designed solo within a team project · 2025 (released as a module after the project ended)

- Reusable unit system with interfaces and ScriptableObject presets: class-based FSM, status effects, per-part hit detection, boss-phase control, Animator-free animation controller

Other Work Projects (Cemware)

- **Pump Simulation Software** (client delivery, team of 2): Implemented all UI interaction and simulation logic; extended UI base classes and used the Observer pattern to add features
- **ImmersionSim** (AR/VR): Built AR Foundation / VRTK / MRTK2 content; WebGL ↔ JavaScript interop for save/load and UGC support

Education · Certifications · Military Service

- Korea National Open University, B.S. in Computer Science (transfer) · Aug 2025 – Aug 2027 (expected)
- Jangan University, Associate Degree in Game Contents · Mar 2019 – Feb 2022
- Certifications: Engineer Information Processing (HRD Korea, Sep 2026), ISTQB CTFL (Sep 2019)
- Military service completed as industrial technical personnel (Nov 2022 – Oct 2024) · Languages: Korean (native), English (technical reading)